TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*High-speed laser grooving of semiconductor wafers with sub-µm roughness for advanced semiconductor packaging.*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials		*Electrics & Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Precision manufacturing machinery	Optical energy-applied processing machines (Code No. 100103)	Machines using optical technology	Laser beam machines (Code No. 200102)
Concept of Proposed Technology	In the advanced semiconductor packaging (AVP) process, the aim of this project is to minimize thermal damage during the fine-line laser grooving process used to separate silicon wafer chips. This can be achieved by developing integrated, ultra-precise and high-quality laser processing technologies and mass-production systems, combining ultrashort pulse laser processing with additional optical and processing technologies.			
Rationale for Joint R&D Project	In current semiconductor manufacturing processes, plans are in place to reduce the width of the dicing street from around 60 µm to approximately 30 µm in order to increase area-based production efficiency. However, as the width of the dicing street decreases, damage caused by thermal deformation becomes a significant issue, rendering conventional grooving processes using nanosecond or picosecond lasers unsuitable. Grooving technology based on ultrashort pulse lasers can minimize thermal deformation and enable localized, high-energy laser irradiation with high			

	productivity. While Korea has world-leading technology and expertise in ultrafast laser processing for display markets, the core technologies for laser sources and processing system components originate from Europe, the USA or Japan. To open new frontiers for laser technologies in advanced packaging, it is necessary to combine Korean technological expertise and access to main end-users with Fraunhofer's extensive know-how in designing and sustaining high-performance UV ultrafast lasers, laser optics, and laser nanostructuring technologies, as well as in minimizing heat-affected zones during laser processing.
Key Words	1. Si/SiC/GaN wafer grooving 2. UV, DUV ultrashort pulse laser 3. Sub-micro roughness 4. DAZ free 5. Flexible beam shaping

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	This project aims to develop integrated ultra-precise and high-quality laser processing technologies and mass-production systems by combining ultrashort pulse laser processing with additional optical and processing technologies. To achieve this, the development is structured into three key areas: process technology development, optical module development, and system development, with the following detailed objectives and tasks: • Development of high-quality grooving process technology using ultrashort pulse lasers and processing techniques to suppress DAZ (Defects Affected Zone) • Development of optical modules and systems for ultra-fine/high-quality processing, including beam-shaping optical systems adaptable to various dicing street widths • Corresponding process development for the variety of targeted materials • Realization of the prototype grooving system optimized for high-efficiency high-throughput processing, and reliability.			
R&D Target Specifications	Prototype-level laser grooving machine with the following key performance indicators (KPI): • Minimum grooving line width: 20 µm • Maximum grooving line width: 100 µm • Grooving DAZ (Damage Affected Zone) size: < 5 µm • Processing depth resolution ≤ 250 nm • Grooving speed: > 150 mm/s • Aspect ratio ≥ 5 • Materials: Si, SiC, and GaN wafers—targeting memory-device packaging and extension into power-device applications.			
R&D Plan	**Phase 1** serves as an evaluation phase, during which mutual visits and online discussions will be conducted to gain a clear understanding of feasible solutions. This will enable a more concrete approach to the goals and development plan of Phase 2. In **Phase 2** technological developments to achieve target specifications will be performed. The project can be organized in 3 work packages: WP1. Development of basic ultra-fine grooving technologies. Here research on laser-matter interaction will be performed to determine quality and speed limiting mechanisms for different wavelengths and pulse durations. First iteration of the required optical designs will be created and benchmarked in simulations in order to realize required beam size and shape and optical error corrections. The test experiments on the relevant materials will be performed to determine the achievable productivity and quality limits. **Milestones: M1. Selection of laser system (Month 6), M2. Optical design within specifications (M12).** WP2. Development of advanced semiconductor packaging process system Here the results of WP1 will be used to design and realize a prototype system. Main tasks will include definition of system requirements, selection and tests of required hardware components, design and construction, integration and software development, and validation of achieved performance. **Milestones: M3. Optical module realized and tested. (M18) M4. Full system assembly completed (M24).** WP3. Process Development, verification and demonstration. In this work package processing of micro and nanometer-level structures in semiconductor materials will be performed, productivity will be evaluated using the system developed in WP2, and the developed process will be demonstrated on relevant end-user wafers with different materials **Milestones: M6. In-spec performance validated on a customer's wafer.**			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer ILT, experts in laser-material processing technology, will use a variety of high-precision laser material microstructuring tools, as well as in-house metrology equipment (SEM, AFM, LSM and WIM), to study the physical limits of the achievable efficiency and quality of laser processes. Fraunhofer ILT will be contributing scientifical and technological know-how to all planned WPs: 1. On the topic of **optical system design** ILT will tackle the tasks of optimizing the spot size and the multi-spot configuration that is essential to meet the project's throughput goals. A larger spot increases throughput but reduces precision; the available average power and beam quality will determine the feasible number of beams in the multi-beam configuration. ILT's modeling and design capabilities [1] will then be used to design and simulate optical system design and phase masks suitable either for tests with spatial light modulators or for manufacturing diffractive optical elements. 2. ILT operates a large number of ultra-short-pulsed (USP) laser systems from IR to DUV wavelengths. For the task of **USP laser selection and benchmarking** these will be used for preliminary characterization and tests. The obtained data will guide selection of the most suitable laser system for the prototype tool - one that is both industrially mature and commercially available. ILT will analyze the throughput–quality trade-off in detail to derive the final recommendation for the prototype tool design. 3. **Defect-free USP material processing** is the core area of Fraunhofer ILT's contribution and will receive the most development effort. Given the broad variety of materials found on production-level semiconductor wafers, it is challenging to establish a laser process that delivers high-quality grooves at high throughput. New approaches must be developed to advance system performance beyond the state of the art. ILT is a leader in ultrashort-pulse processing of various materials at wavelengths from DUV to MIR and will contribute its extensive scientific know-how [2] to the development of high-quality grooving. 4. ILT will provide Korean partners with access to **EU and German networks** for laser-assisted manufacturing and laser producers. 5. Fraunhofer ILT will provide its background IP the project partners for the duration of the project. Developing successful market-ready technology will enable Korean partners to obtain rights to Fraunhofer foreground **IP** on preferential conditions
Expected R&R of Korean SME Partner	1. Korean SME leading the project would have rich experience in building integrated laser-based setups for microelectronic and display industry. It would contribute to the project by integrating the research of ILT into the demonstrator and projecting the state-of-the-art knowledge and technology requirements of the end user. 2. The SME will also ensure that ILT is provided with relevant test samples and the technical requirements of end users for the developed process. 3. The SME will lead commercialization efforts encompassing regulatory compliance, IP management, and market strategies.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Prototype-level laser grooving machine with industrially relevant performance that will enable Korean SME to access growing >$11 billion market for wafer grooving tools • Generation of new IP (patents/invention) for both partners • Joint publications and presentations at international conferences and fairs • Follow-up of publicly funded or bilateral projects aiming at full commercialization of the technology (reaching TRL 8-9)

6. References

[1] USP Multibeam Technology of Fraunhofer ILT: https://www.youtube.com/watch?v=KMoeun2aH9w

[2] S. Danylyuk et al., Highly efficient UV and DUV laser micro- and nanoprocessing for microelectronics applications. LASE (2025) https://doi.org/10.1117/12.3042723

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization